

EXPERIMENT-1

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Date: 08/07/2026
Subject Code: 24CSH-337

AIM:

Understand the concepts of :

- (i) IP Address,
- (ii) Cisco IOS,
- (iii) Straight & Cross Cable,
- (iv) RJ45,
- (v) Layer 2 Switch,
- (vi) Router

and to simulate their working using Cisco Packet Tracer.

OBJECTIVES:

- To understand IP addressing and subnet masks.
- To study the Cisco IOS Command Line Interface (CLI).
- To understand Straight-through and Cross-over cables.
- To study the RJ-45 connector and wiring standards.
- To understand the working of a Layer-2 Switch.
- To understand the role of a Router in connecting different networks.
- To verify communication using the Ping command.

SOFTWARE REQUIREMENTS:

(a) Software:

- Cisco Packet Tracer
- Windows Operating System

(b) Hardware (Virtual):

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- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

THEORY: (i) IP Address

An IP (Internet Protocol) Address is a unique logical address assigned to every device connected to a network. It identifies the sender and receiver during communication, allowing data to reach the correct destination. IPv4 uses a 32-bit address represented in dotted decimal notation (e.g., 192.168.1.10), whereas IPv6 uses a 128-bit address to provide a much larger address space. A subnet mask divides an IP address into the Network ID and Host ID, where the Network ID identifies the network and the Host ID identifies the specific device within that network. During communication, the operating system places the source and destination IP addresses in the IP packet header, and networking devices forward the packet until it reaches the destination.

(ii) Cisco IOS

Cisco IOS (Internetwork Operating System) is the operating system used in Cisco routers and switches. It provides a Command-Line Interface (CLI) that allows administrators to configure, monitor, and manage networking devices. Cisco IOS is used to assign IP addresses, configure routing, create VLANs, set passwords, monitor interfaces, and save device configurations. It has three main operating modes: User EXEC Mode, Privileged EXEC Mode, and Global Configuration Mode, each providing different levels of access. Common commands include enable, configure terminal, show running-config, show ip interface brief, hostname, and copy running-config startup-config.

(iii) Straight Cable and Cross Cable

Straight-through and Cross-over cables are Ethernet cables used for connecting networking devices. A Straight Cable is used to connect different types of devices, such as PC to Switch and Switch to Router, with both ends following the same wiring standard (T568A-T568A or T568B-T568B). A Cross Cable is used to connect similar devices, such as PC to PC and Switch

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to Switch, with one end using T568A and the other using T568B so that the transmit and receive wires are crossed. Modern devices support Auto-MDIX, which automatically detects the required wiring, making either cable usable in many cases.

(iv) RJ-45 Connector

The RJ-45 connector is the standard connector used with Ethernet cables in computer networks. It contains 8 pins arranged according to the T568A or T568B wiring standards. The connector is used to terminate twisted-pair cables and establish wired network communication between devices. Straight-through cables use the same standard at both ends, whereas cross-over cables use different standards at each end.

(v) Layer-2 Switch

A Layer-2 Switch operates at the Data Link Layer (Layer 2) of the OSI model. It forwards Ethernet frames using MAC addresses, allowing devices within the same Local Area Network (LAN) to communicate efficiently. The switch maintains a MAC address table and sends frames only to the required destination port, reducing unnecessary network traffic.

(vi) Router

A Router is a Layer-3 networking device that connects different networks and forwards packets based on IP addresses. It maintains a routing table to determine the best path for data transmission. Routers separate broadcast domains and enable communication between different LANs and external networks such as the Internet.

STEPS/IMPLEMENTATION:

Part I – IP Address

1. Place two PCs and one Layer-2 Switch.
2. Connect both PCs using Copper Straight-Through cables.
3. Assign IP addresses.

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4. Verify connectivity using the Ping command.

Part II – Cisco IOS

5. Add a Cisco 1941 Router.
6. Open the CLI.
7. Explore User EXEC, Privileged EXEC, and Global Configuration modes.
8. Execute basic Cisco IOS commands.

Part III – Straight Cable & Cross Cable

9. Demonstrate Straight-through cable connections.
10. Demonstrate Cross-over cable connections.
11. Observe the difference between both cable types.

Part IV – RJ-45 Connector

12. Study the RJ-45 connector and wiring standards.

Part V – Layer-2 Switch

13. Connect multiple PCs to a Layer-2 Switch.
14. Verify communication using Ping.

Part VI – Router

15. Connect two different networks using a router.
16. Configure router interfaces.
17. Verify communication between both networks.

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PROCEDURE:

1. IP Address

- Open Cisco Packet Tracer.
- Add two PCs and one switch.
- Connect them using Copper Straight-Through cable.
- Assign IP addresses:
 - PC0 – 192.168.1.10 / 255.255.255.0
 - PC1 – 192.168.1.20 / 255.255.255.0
- Verify communication using:
ping 192.168.1.20

2. Cisco IOS

- enable
- show running-config
- show startup-config
- show ip interface brief
- configure terminal
- hostname R1
- copy running-config startup-config

3. Straight Cable

- Connect PC → Switch.
- Connect Switch → Router.

Cross Cable

- Connect PC → PC.
- Connect Switch → Switch.

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4. Layer-2 Switch

- Connect three PCs to a switch.
- Assign IP addresses.
- Verify communication using Ping.

5. Router

- Connect two switches to a router.
- Configure router interfaces.
- Configure default gateways on PCs.
- Verify communication using Ping.

INPUT/OUTPUT DETAILS:

(a) Devices Used

- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

(b) Cable Types

- Copper Straight-Through
- Copper Cross-Over

(c) Cisco IOS Commands Used

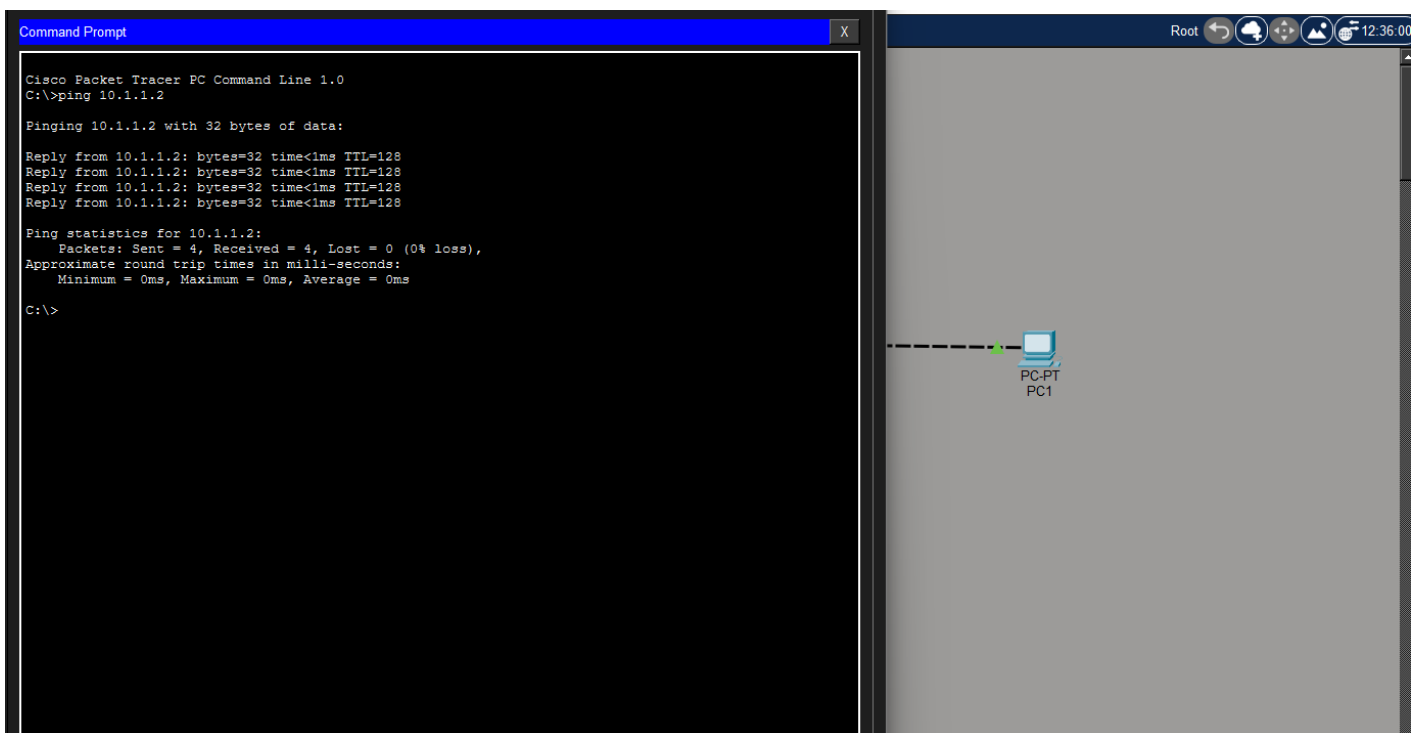
- enable
- configure terminal
- hostname R1

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- show running-config
- show startup-config
- show ip interface brief
- copy running-config startup-config
- ping

DEMONSTRATION/OUTPUT:

Part-1: IP Address



The screenshot displays the Cisco Packet Tracer interface. On the left, a Command Prompt window titled "Command Prompt" shows the output of a ping command. The text in the window is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

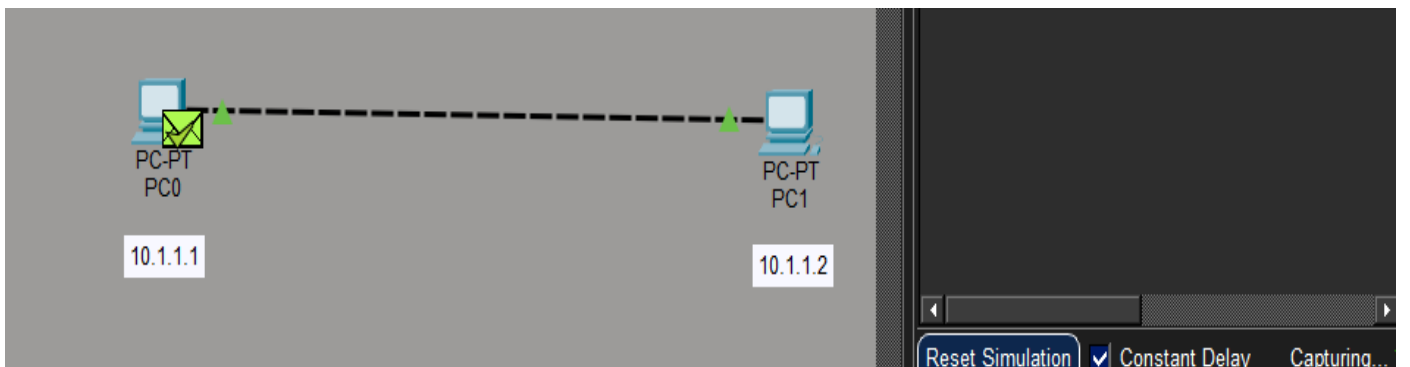
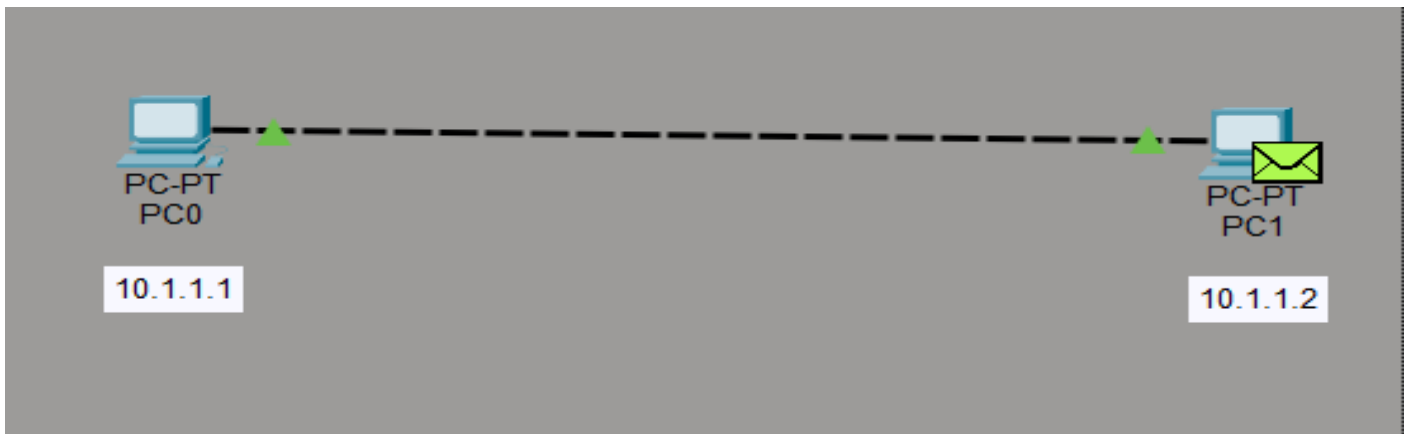
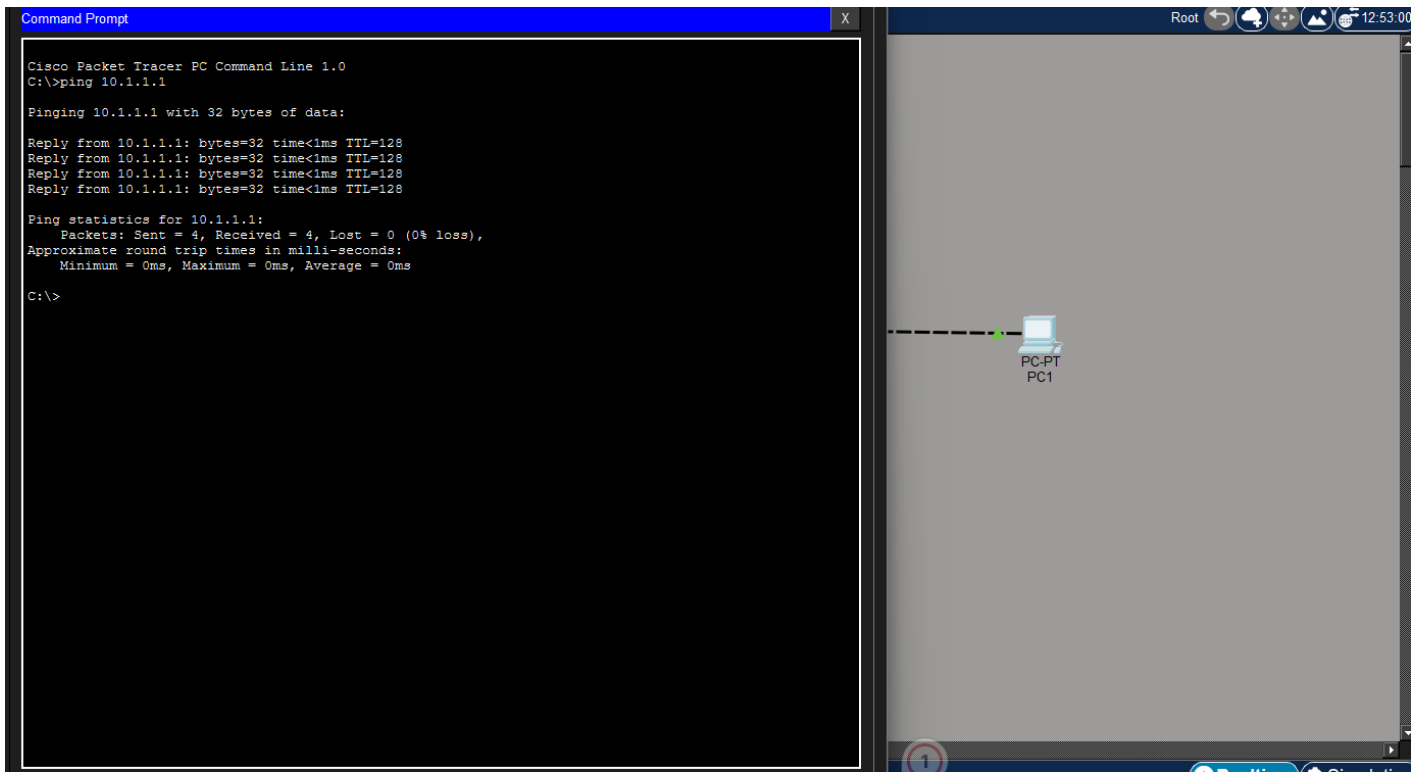
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

On the right, the network diagram shows a single device labeled "PC-PT PC1" connected to a network.



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Part-2: Cisco IOS

(a) User EXEC Mode (Router>)

```

ipve      Global IPv6 Configuration Commands
key       Key management
license   Configure license features
line     Configure a terminal line
lldp     Global LLDP configuration subcommands
logging   Modify message logging facilities
login     Enable secure login checking
mac-address-table  Configure the MAC address table
no        Negate a command or set its defaults
ntp       Configure NTP
parser    Configure parser
policy-map  Configure QoS Policy Map
port-channel  EtherChannel configuration
priority-list  Build a priority list
privilege  Command privilege parameters
queue-list  Build a custom queue list
router    Enable a routing process
secure    Secure image and configuration archival commands
security  Infra Security CLIs
service   Modify use of network based services
snmp-server  Modify SNMP engine parameters
spanning-tree  Spanning Tree Subsystem
tacacs-server  Modify TACACS query parameters
username  Establish User Name Authentication
vpdn      Virtual Private Dialup Network
vpdn-group  VPDN group configuration
Router(config)#
Router(config)#
Router(config)#

```

Router0

Physical Config **CLI** Attributes

```

Router#disable
Router>show clock
*0:6:12.272 UTC Mon Mar 1 1993
Router>show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 6 minutes, 14 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wml/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:
License UDI:

-----
Device# PID SN
-----
*0 CISC02911/K9 FTX1524C282-

Technology Package License Information for Module:'c2900'
-----
Technology Technology-package Technology-package
Current Type Next reboot
-----
ipbase ipbasek9 Permanent ipbasek9
security None None None

```

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```
Router>
Router>ping 127.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 127.0.0.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router>enable
Router#show running-config
Building configuration...

Current configuration : 693 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524CZ82-
!
!
!
!
!
!
!
!
```

```
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

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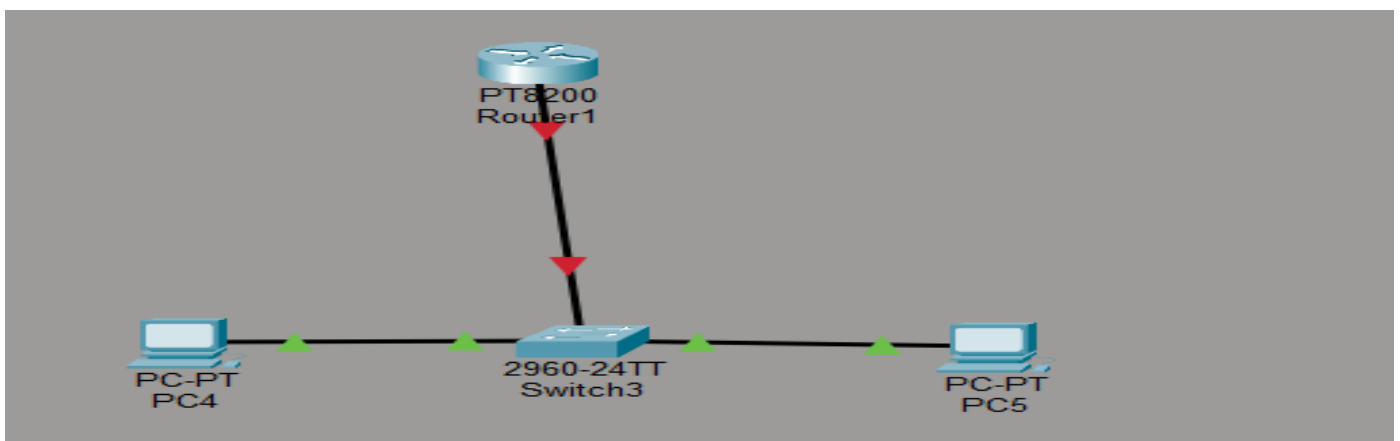
(b) Privileged EXEC Mode(Router#)

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname AITRouter
AITRouter(config)#
AITRouter(config)#
AITRouter(config)#
AITRouter(config)#hostname Router
Router(config)#enable secret 123
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console
^Z
Router#disable
Router>enable
Password:
Password:
Router#
```

(c) Global Configuration Mode(Router(config)#)

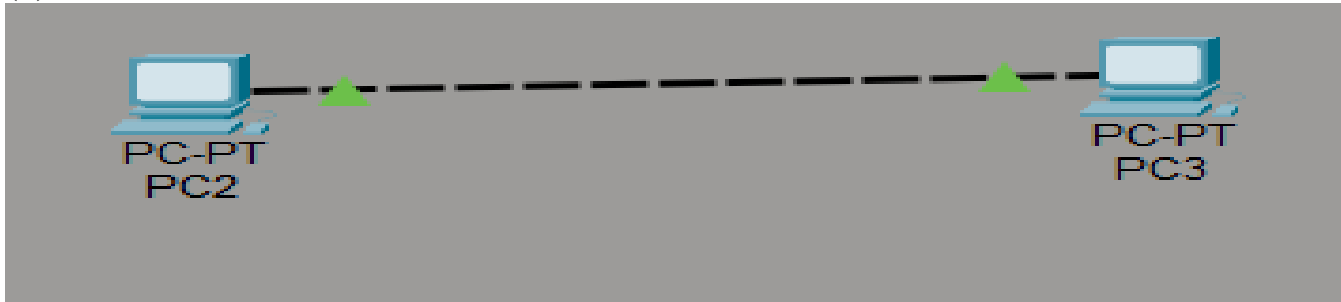
```
show history
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#banner motd #Welcome#
R1(config)#enable secret 123
R1(config)#service password-encryption
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

Part-3: (a) Straight Cable



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(b) Cross Cable



Part-4: Layer-2 Switch



Part-5: Router



LEARNING OUTCOMES:

After completing this experiment:

- I understood the fundamentals of IP addressing and subnetting by identifying the Network ID and Host ID, assigning valid IPv4 addresses, and verifying communication between devices using the Ping command.
- Gained practical knowledge of Cisco IOS and its Command-Line Interface (CLI) by navigating different operating modes (User EXEC, Privileged EXEC, and Global Configuration), executing basic commands, and performing initial router configuration.

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- Differentiated between Straight-through and Cross-over Ethernet cables by understanding their wiring standards (T568A and T568B), identifying their applications, and selecting the appropriate cable for connecting similar and different networking devices.
- Understood the functions of RJ-45 connectors, Layer-2 switches, and routers by studying their roles in wired communication, frame forwarding using MAC addresses, and packet routing between different IP networks.
- Developed the ability to design, configure, and troubleshoot basic network topologies in Cisco Packet Tracer by connecting network devices, assigning IP addresses, configuring routers and switches, and testing end-to-end connectivity successfully.